Amagi Framework v1.0

Architecture as Compliance — A Design-Centric Approach to EU AI Act Conformity

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Abstract

The EU AI Act mandates that providers of high-risk AI systems demonstrate **structural, auditable, and continuous compliance** across risk management, data governance, human oversight, and post-market monitoring.

Current industry approaches rely on **post-hoc documentation and procedural controls** — resulting in audit failures, compliance gaps, and legal exposure.

Amagi Framework v1.0 introduces a paradigm shift: compliance by architectural construction.

This is not software, service, or library — but a reference architecture that structurally enforces:

☑ Human oversight — systems cannot operate without designated control

☑ Complete auditability — immutable, regulator-ready decision trails

☑ Inherent data governance — bias, drift, and quality validation at ingress

☑ Constrained communication — only pre-approved interactions permitted

☑ Controlled updates — no automatic or silent changes allowed

☑ Automatic incident containment — systems enter safe state without delay

This architecture enables presumption of conformity with EU AI Act Annex III requirements.

Implementation responsibility lies with the user; the author assumes no liability.

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1. Introduction

Organizations face a critical challenge:

How to demonstrably prove AI system safety — continuously, under all conditions?

Regulators no longer accept **promises or documentation alone**. They demand **structural proof**.

Amagi Framework v1.0 provides this proof — not through policies, but through **architectural design**.

It doesn't merely suggest how safety might be achieved.

It demonstrates that safety is structurally guaranteed.

This document outlines the **six Core Principles** of the framework — while protecting implementation-sensitive elements.

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2. The Six Core Principles

Amagi Framework v1.0 is defined by six **Core Principles**, which must be implemented together as an indivisible mechanism for structural compliance:

1. **Immutability** — Policies, logs, and access rules are cryptographically signed and unmodifiable after deployment.
2. **Constrained Communication** — All inter-component data flows are strictly governed by a declarative, signed graph; non-listed flows are blocked at the network level.
3. **Contextual Oversight** — Critical decisions require validated human approval (DHO) based on risk evaluation and source context.
4. **Controlled Evolution** — All updates, configuration changes, or model swaps require manual approval and audit trail; no silent or automatic changes.
5. **Automatic Safety Preservation** — Upon detection of failure, drift, or policy violation, the system enters a safe, contained state automatically, without human intervention.
6. **Complete Audit Integrity** — All decisions, interventions, and system states are logged in an immutable, dual-stored, cryptographically signed trail.

These principles are enforced **architecturally**, not procedurally.

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3. Legal Foundation: Copyright Protection of Architecture

This work is an **original intellectual creation** protected under:

- **Art. XI.295 of the Belgian Code of Economic Law**:

  "A computer program is protected if it is original, in the sense that it is an intellectual creation of its author."

- **CJEU Infopaq (C-5/08)**:

  Copyright protects the *"intellectual creation specific to its author"*, including the *choice and disposition of structural elements*.

- **Whelan v. Jaslow (3rd Cir. 1986)**:

  Protection extends to the **Structure, Sequence, and Organization (SSO)** of a system, even beyond literal code.

- **SPF Économie / Copyright.eu**:

  "Software architecture, structure and organization are protected... The simple adoption of the architecture of one software for another is likely to constitute a counterfeit."

The Amagi Framework v1.0 qualifies as **preparatory design material** (flow charts, policy definitions, communication graphs, system organization) — explicitly protected under EU and Belgian law.

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4. Derivative Architecture and Structural Imitation

4.1 Derivative Architecture

Any system — regardless of naming, terminology, programming language, or implementation technology — that implements **all six Core Principles** defined in Section 2 constitutes a **Derivative Architecture of Amagi Framework v1.0**.

This includes systems using alternative component names (e.g., “Safety Interceptor”, “Policy Engine”) **if functional behavior, interaction pattern, and constraints match the six guarantees**.

4.2 Structural Imitation

Any system that adopts the **core structural pattern** — specifically:

- A centralized control plane governing all inter-component communication,
- A declarative, signed graph defining permitted data flows,
- Mandatory human approval gates triggered by risk evaluation,
- Immutable, dual-channel logging of all decisions and interventions,
- Architectural containment activated without human intervention,

— constitutes **Structural Imitation**, **even if only a subset of the six guarantees is implemented**, additional modules are added, or components are renamed.

Such systems **derive their compliance logic from the Amagi architectural blueprint** and therefore **require licensing under S-APL** for commercial use.

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5. Business and Regulatory Value

5.1 For Regulators & Auditors

- Provides a **reference architecture for presumption of conformity**
- Enables **predictable, efficient audits** based on structural proof

5.2 For Enterprise Organizations

- Reduces audit time by 60–80%
- Eliminates custom compliance engineering
- Avoids potential fines up to 6% global turnover

5.3 For Technology Partners

- Creates opportunity to build **Amagi-compliant solutions** under license

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6. Implementation Path

1. **Execute NDA** to review full architectural specifications
2. **License the framework** under S-APL terms
3. **Implement** with your engineering team using provided templates
4. **Achieve certification** through remote compliance audit

Initial licensing priority will be given to organizations in regulated sectors with immediate EU AI Act compliance requirements.

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7. Conclusion

Amagi Framework v1.0 transforms AI safety from **retrospective verification** to **architectural guarantee**.

It doesn't ask: "Can you prove your system is safe?"

It ensures: "Your system cannot be unsafe."

This is not merely another tool.

It represents a new foundation for trustworthy artificial intelligence.

We invite regulators, auditors, and industry leaders to recognize Amagi Framework v1.0 as the reference architecture for EU AI Act conformity.

References

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Amagi Framework v1.0 — Architectural Guarantees for AI Safety & Compliance